Sam Kim Miller

Curriculum Vitae

Department of Mathematics
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Mathematics Employment and Education

2025–Present **Limited-Term Assistant Professor**, *University of Georgia*
Mentors: Nate Harman, Dan Nakano

2019–2025 **Ph.D. in Mathematics**, *University of California, Santa Cruz*
Advisor: Robert Boltje

2013–2017 **B.S. in Mathematics**, *Harvey Mudd College*
Advisors: Arthur Benjamin, Mohamed Omar

Grants, Honors, and Awards

2025, 2026 US Junior Oberwolfach Fellow

2025 UCSC Department of Mathematics (Inaugural) Outstanding Research Award

2025–Present Sigma Xi Full Membership

2025–2027 AMS-Simons Travel Grant

2024–2025 UCSC Chancellor's Dissertation Fellowship, Full-Year

2014–2016 Putnam Top 500 Finishers

2013 ICTM State Math Contest Individual Winner

2013 USCF SuperNationals: 1st Place (U1200 Division)

Research

Interests:
Representation theory, tensor-triangular geometry, permutation modules, finite(ish) groups

In preparation and ongoing projects

The noncommutative tensor triangular geometry of group algebras, *with Kent Vashaw*

Local Picard groups for permutation modules

Periodicity in the stable permutation category, *with Martin Gallauer*

Papers and preprints

1. **Challenging knight's tours**, *with Arthur T. Benjamin (undergraduate research)*
   Math Horizons, 25 (3), 18-21 (2018)
2. **A proof of the optimal leapfrogging conjecture**, *with Arthur T. Benjamin (undergraduate research)*
   Involve, 18 (1), 105–122 (2025), arXiv:2110.08319
3. **Endotrivial complexes**
   J. Algebra, 650, 173-218 (2024), arXiv:2309.12138

4. **Brauer pairs for splendid Rickard equivalences**, *with Jadyn V. Breland*
   J. Algebra, 691, 694-729 (2026), arXiv:2312.10258
5. **Relatively endotrivial complexes**
   J. Pure Appl. Algebra, 229 (2), 107867 (2025), arXiv:2402.08042
6. **The classification of endotrivial complexes**
   Adv. Math. 478, 110414 (2025), arXiv:2403.04088
7. **Galois descent of splendid Rickard equivalences between blocks of p-nilpotent groups**
   Proc. Amer. Math. Soc. 153 (5), 1893-1902 (2025), arXiv:2405.16061
8. **On endosplit p-permutation resolutions and Broué's conjecture for p-solvable groups**
   Represent. Theory 30, 151-174 (2026), arXiv:2408.04094
9. **On functoriality and the tensor product property in noncommutative tensor-triangular geometry**
   Submitted, arXiv:2505.01899
10. **The Euler characteristic of an endotrivial complex**, *with Nadia Mazza*
    J. Pure Appl. Algebra, 230 (10, Special Section in honor of Daniel K. Nakano's 60th birthday), 108345 (2026), arXiv:2508.07404
11. **Permutation twisted cohomology, remixed**
    Submitted, arXiv:2509.00954
12. **Re-framing the classification of ideals in noncommutative tensor-triangular geometry**, *with Timothy De Deyn*
    To appear, Adv. Math., arXiv:2510.23767
13. **A semisimple subcategory of Khovanov's Heisenberg category**
    Submitted, arXiv:2512.13968
14. **The classification of integral endotrivial complexes**, *with Juan Omar Gómez*
    Submitted, arXiv:2605.31128
15. **The fusion-stable tom Dieck homomorphism**
    Submitted, arXiv:2608.12499
16. **Non-orientable representation spheres**
    arXiv:2608.18015
17. **Appendix to: Periods in equivariant and motivic contexts by Martin Gallauer**, *with Martin Gallauer*
    Submitted, arXiv:2511.14325
18. **The tensor-triangular geometry of some Delannoy categories**, *with Nate Harman*
    Draft available upon request
19. **Permutation twisted cohomology, reduced**, *with Juan Omar Gómez*
    Draft available upon request

Proceedings

1. **Endotrivial complexes and remixed twisted cohomology**
   Oberwolfach Reports 23 (1), 2026

2. **Endotrivial complexes and remixed twisted cohomology**
OCAMI Reports, to appear

Theses

B.S. **The polynomial combinatorial Hirsch conjecture**, *2017*
Advisor: Mohamed Omar

Ph.D. **Permutation modules and endotrivial complexes**, *2025*
Advisor: Robert Boltje

Talks

Expository and Outreach

??? 2027 Carleton College Math Colloquium
Apr 2024 University of San Francisco Math Colloquium
Oct 2022 Science on Tap hosted by Society of Women Engineers, UC Santa Cruz chapter
Feb 2021 University of California, Santa Cruz Mathematics Graduate Student Seminar

Research

??? 2027 University of Utah Algebra Seminar
??? 2027 University of Washington Algebra Seminar
Mar 2027 University of Oregon Algebra Seminar
Dec 2026 Harvard University Babytop Seminar
Nov 2026 AMS Western Sectional Special Session
Oct 2026 Johns Hopkins University Topology Seminar
Oct 2026 Rice University Algebra Seminar
Jun 2026 University of Warwick Algebra Seminar (cancelled due to illness)
Jun 2026 Aarhus University Algebra Seminar
May 2026 University of Bielefeld BIREP Oberseminar
May 2026 UC Irvine Algebra Seminar
May 2026 UC San Diego Algebra Seminar
Mar 2026 AMS Southestern Sectional Special Session
Feb 2026 Oberwolfach Workshop 2609: Cohomology of Finite Groups
Dec 2025 Spectrums in Representation Theory of Algebras and Related Topics
Nov 2025 UCLA Algebra Seminar
Oct 2025 University of California, Santa Cruz Algebra and Number Theory Seminar
Sep 2025 UGA Algebra Seminar (x2)
Aug 2025 AMS Western Sectional Special Session
Aug 2025 UGA Graduate Student Seminar
May 2025 New Directions in Group Theory and Triangulated Categories
Apr 2025 AMS Spring Northeastern Sectional Special Session
Jan 2025 University of Washington Algebra Seminar
Dec 2024 University of Copenhagen Topology Seminar

Nov 2024	University of California, Berkeley Representation Theory and Tensor Categories Seminar
Nov 2024	University of California, Santa Cruz Algebra and Number Theory Seminar
Oct 2024	AMS Western Sectional Special Session
Oct 2024	University of California, Santa Cruz Mathematics Graduate Student Seminar
Oct 2024	AMS Southeastern Sectional Special Session
Jul 2024	Group Actions, Groups, and Geometry
Jun 2024	Groups and Algebras in Bicocca for Young Algebraists 2024
Jun 2024	RPTU Kaiserslautern-Landau Algebra Seminar
Jun 2024	Groups and their Actions: Algebraic, Geometric, and Combinatorial Aspects
May 2024	Representation Theory and Related Geometry: Progress and Prospects (Nakano 60)
May 2024	UCLA Algebra Seminar
May 2024	University of California, Santa Cruz Algebra and Number Theory Seminar
May 2024	AMS Western Sectional Special Session
Jan 2024	University of Arizona Algebra Seminar
Jan 2024	Joint Mathematics Meetings Contributed Paper Session
Dec 2023	University of Bilkent Algebra Seminar
Nov 2023	University of Georgia Algebra Seminar
Oct 2023	University of California, Santa Cruz Mathematics Graduate Student Seminar
Sep 2023	Algebra and Number Theory in Conversation at Manchester
May 2023	AMS Western Sectional Special Session
Jan 2023	Joint Mathematics Meetings Contributed Paper Session
Nov 2022	University of California, Santa Cruz Mathematics Graduate Student Seminar
May 2022	University of California, Santa Cruz Postdoc Symposium
May 2022	University of California, Santa Cruz Mathematics Graduate Student Seminar
Apr 2022	University of California, Santa Cruz Graduate Student Symposium
Aug 2018	MAA MathFest Contributed Session
May 2017	Harvey Mudd Senior Thesis Day
Apr 2017	MAA SoCal/NV Sectional Poster Session

Teaching

At UGA

Spring 2027	Topics in Algebra: Modular Representation Theory of Finite Groups
Fall 2026	Calculus I (x2)
Spring 2026	Precalculus (x2)
Fall 2025	Precalculus (x2)

At UC Santa Cruz: Instructor of Record

Spring 2024	Combinatorics
Winter 2024	Introduction to Number Theory

Fall 2023 Problem Solving Strategies
Summer 2023 Cryptography
Summer 2022 Introduction to Number Theory
Summer 2021 Introduction to Proof and Problem Solving
Spring 2021 Introduction to Partial Differential Equations

At UC Santa Cruz: Teaching Assistant

Spring 2023 Linear Algebra
Winter 2023 Programming for Mathematics
Fall 2022 Advanced Linear Algebra
Spring 2022 Industrial Mathematics
Winter 2022 Programming for Mathematics
Fall 2021 Vector Calculus
Winter 2021 Linear Algebra
Fall 2020 Linear Algebra
Summer 2020 Introduction to Calculus of Several Variables
Spring 2020 Numerical Analysis
Winter 2020 Calculus for Science, Engineering, and Mathematics II
Fall 2019 Calculus for Science, Engineering, and Mathematics I

Organization and Service

Conferences

2026 **AMS Fall Southeast Sectional**, Organizer
Special session on geometric and homological methods in representation theory

2026 **AMS Fall Western Sectional**, Organizer
Special session on finite groups and representation theory

Organization

2025–Present **UGA Algebra Seminar**
Organizer

2022–2024 **University of California, Santa Cruz Mathematics Graduate Student Seminar**
Organizer

2022 **University of California, Santa Cruz Graduate Student Committee**
Electoral Board Member

2021 **University of California, Santa Cruz Mathematics Department Social Committee**
Organizer

Organized Reading Seminars

2025 Introduction to Soergel Bimodules
2025 Introduction to Tensor Categories
2024 Blocks of Cyclic Defect

2021 Local Class Field Theory

2020 Category Theory

Mentorship

2026 University of Georgia Directed Reading Program

2021–2025 University of California, Santa Cruz Peer Mentorship Program

2021, 2024 University of California, Santa Cruz Directed Reading Program

Refereeing and Reviewing

Algebras and Representation Theory

Communications in Algebra

EMS Press

Forum of Mathematics, Sigma

Journal of the London Mathematical Society

Transactions of the American Mathematical Society

MathReviews reviewer

Other Employment

2017–2019 **SupplyFrame**, *Platform Engineer*
Software engineer, worked with big data, natural language processing, and machine learning

2014–2017 **Harvey Mudd College**, *Grutor*
Graded and tutored for various undergraduate courses in mathematics and computer science

2014–2015 **YMCA Camp Widjiwagan**
Led intense week-long canoeing trips for middle- and high-school aged children in the Boundary Waters Canoe Area Wilderness of Minnesota; emphasizing personal growth, self-confidence, and respect for self, nature, others, and equipment